SUSTAINABILITY



Cooling the Cloud: *Regional Tradeoffs in U.S. Data Center Cooling Architectures*

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Introductions



Sammy Houssainy, PhD, PE
Data Center Solutions Architect
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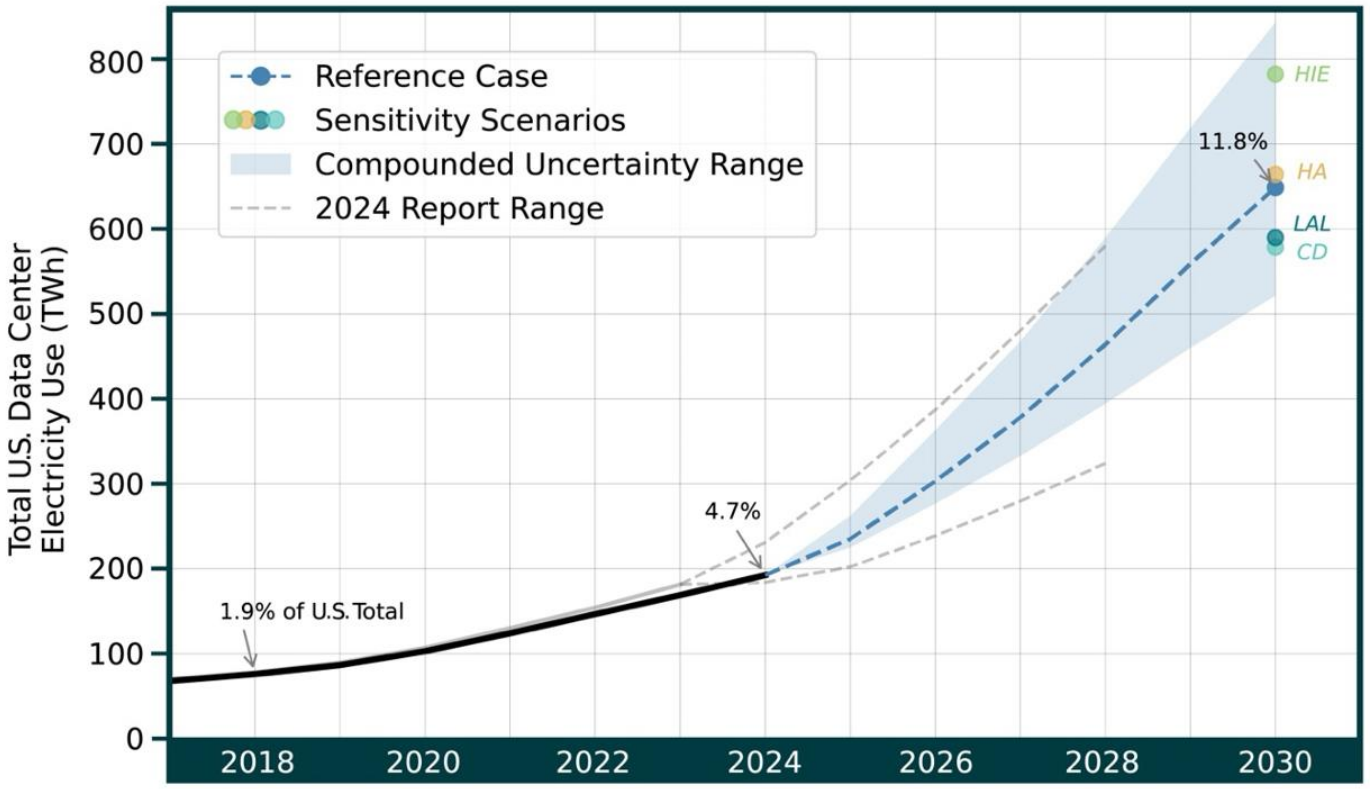


Karim Shalash, PhD
Data Center Solutions Director
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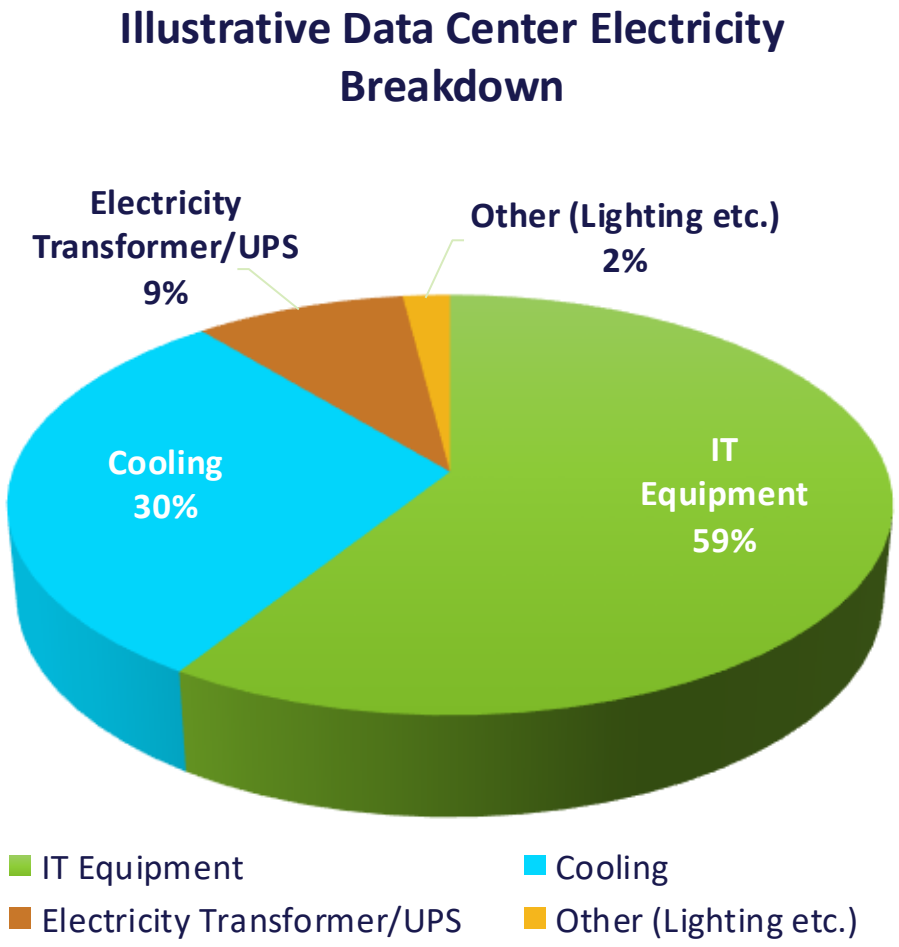
Agenda

- Why cooling selection matters
- Dry-cooled vs. wet-cooled
- Regional modeling approach
- Water-energy selection index
- Carbon implications through 2050
- Takeaways & call to action

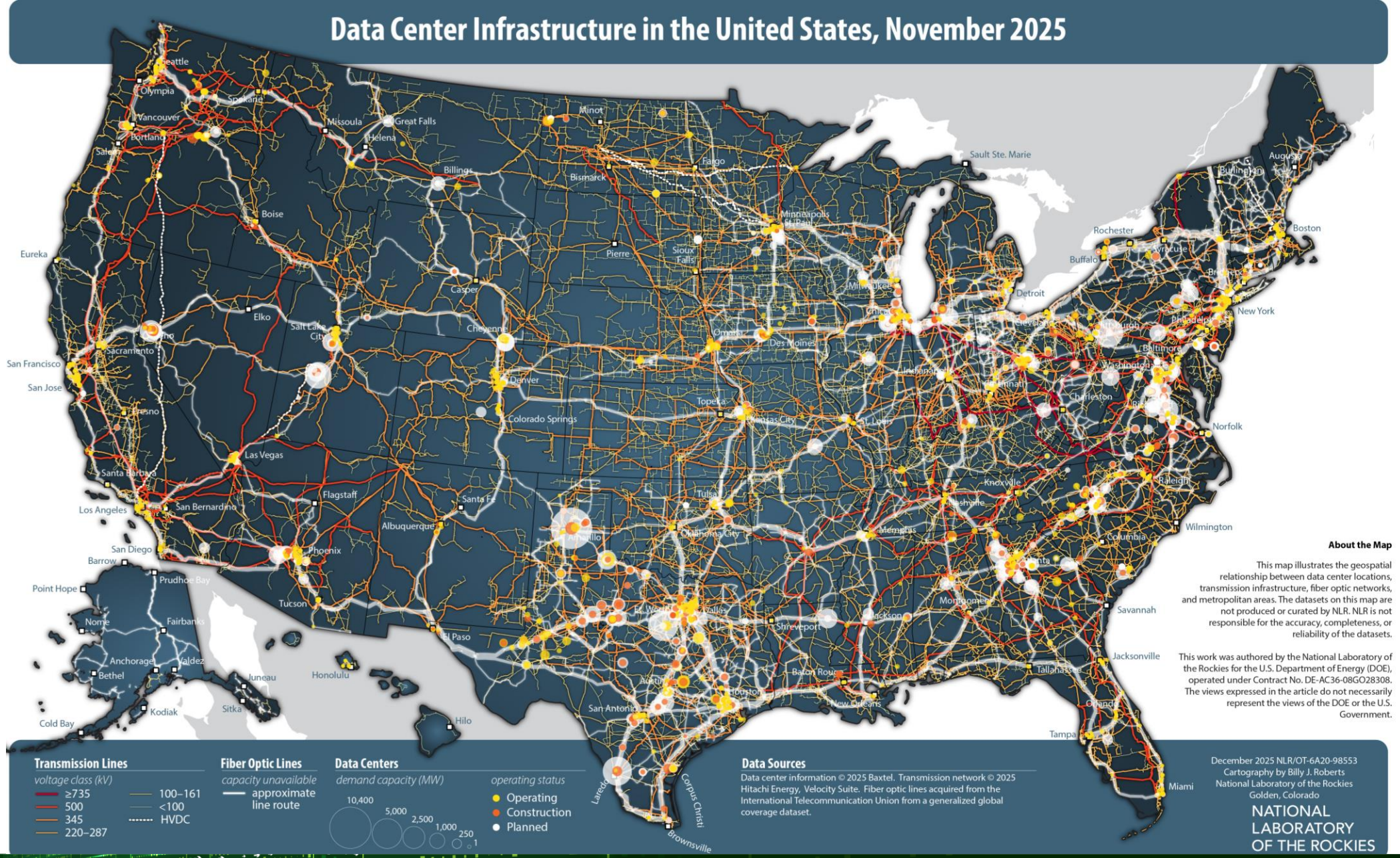
U.S. Data Center Energy Consumption and Breakdown



Sarah J. Smith et all, 2025 United States Data Center Energy Usage Report: 2025 Update, LBNL



Data Centers Are Built Around Power and Fiber Optic Lines ***— Not Cooling Climate***



Dry-Cooled or Wet-Cooled: *Is the Water Worth It?*

Known tradeoff between energy efficiency and water consumption. But what about operating cost and emissions, today and in the future?

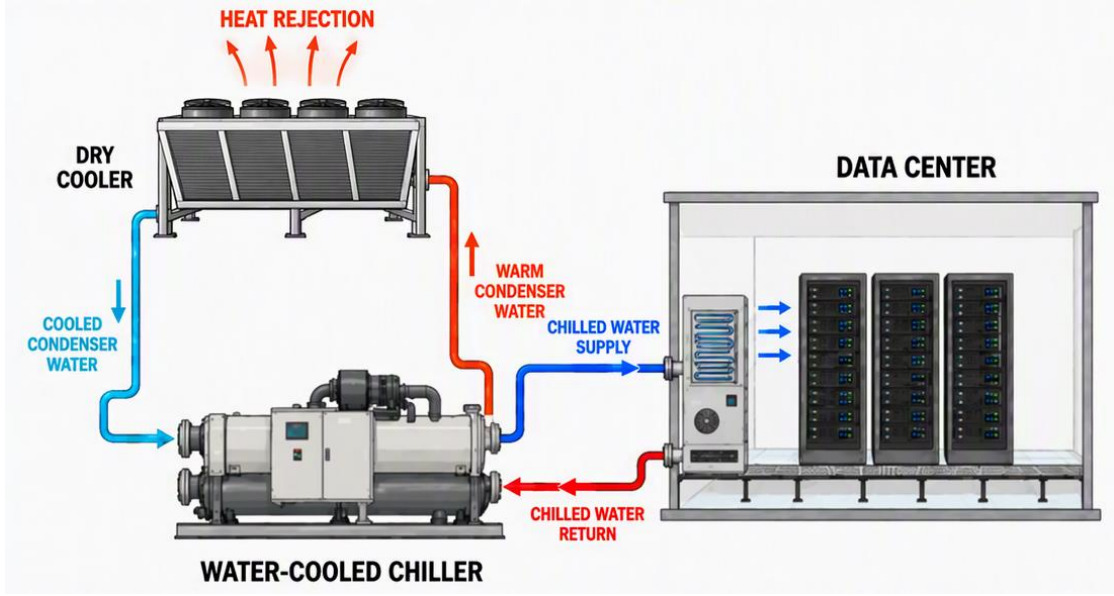
Dry-cooler + water-cooled chiller + economizer

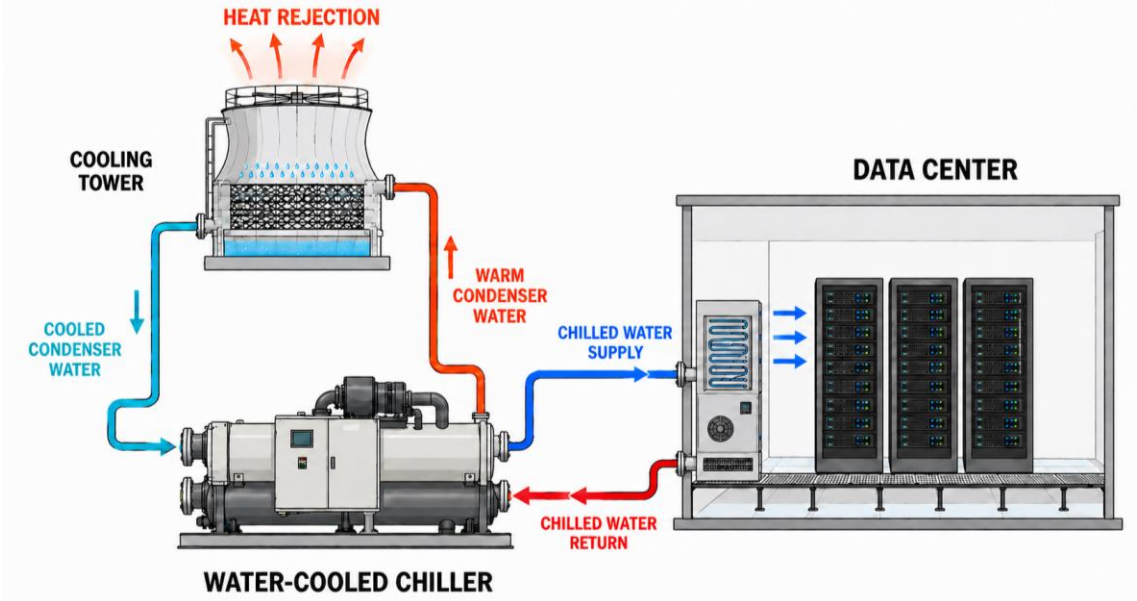
- No cooling-tower water
- Higher condensing temperature
- Generally higher compressor energy

Wet-cooler + water-cooled chiller + economizer

- Lower condenser-water temperature
- Lower chiller energy and peak demand
- Requires continuous evaporative water

Does the value of the electricity saved justify the water consumed?



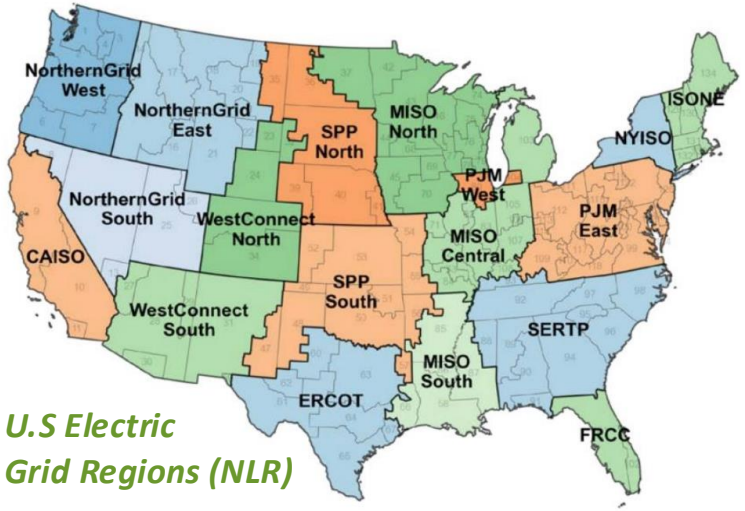


Hourly Regional Comparison

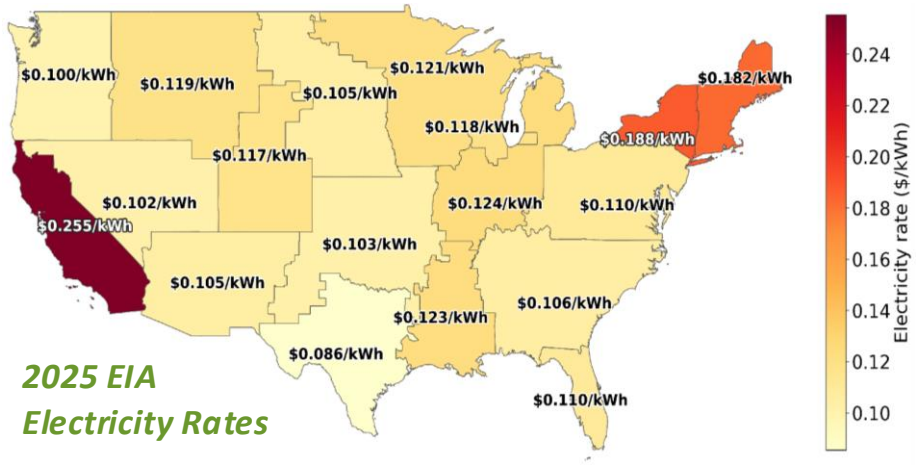
Approach:

- **1 MW-IT load**
- **Dry cooler** + water- cooled chiller + economizer
- **Wet Tower** + water-cooled chiller + economizer
- **Hourly simulations** across 18 U.S. electric-grid regions
- **Regional** weather (TMY3), electricity, demand, and water/wastewater rates
- **2025 eGRID Emissions +** projected hourly NREL Cambium dataset **emission factors through 2050**

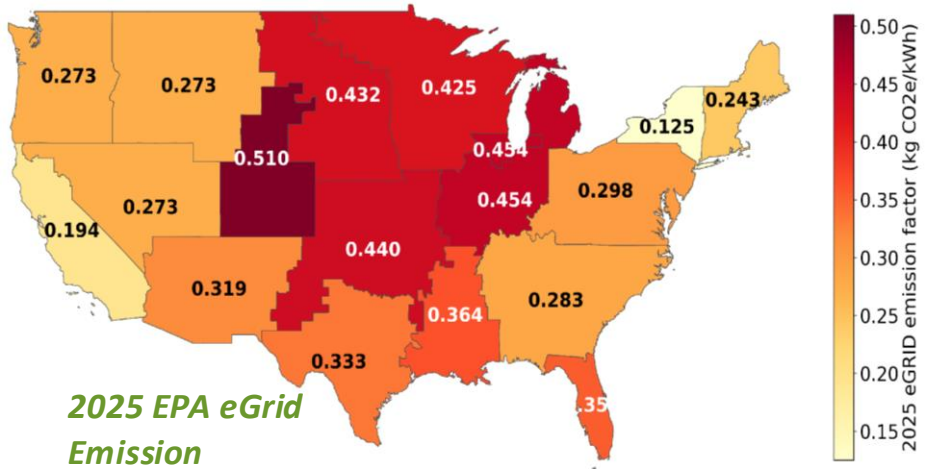
Outputs: annual energy | peak demand | water | operating cost | GHG



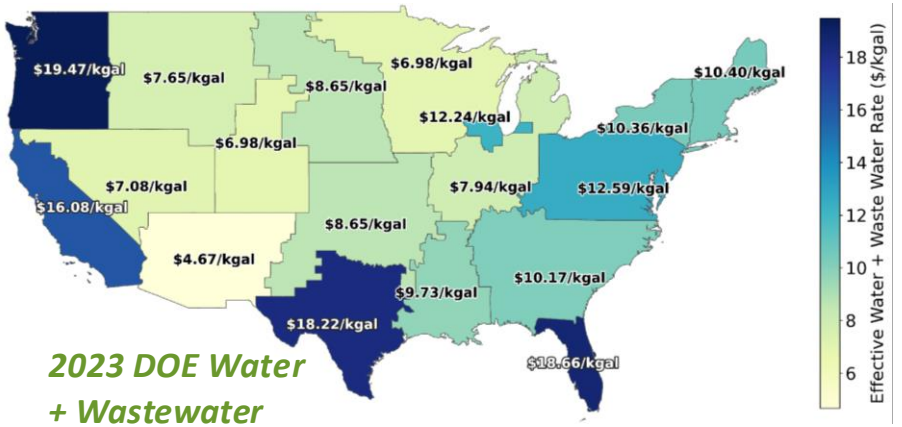
U.S Electric Grid Regions (NLR)



2025 EIA Electricity Rates



2025 EPA eGrid Emission Factors



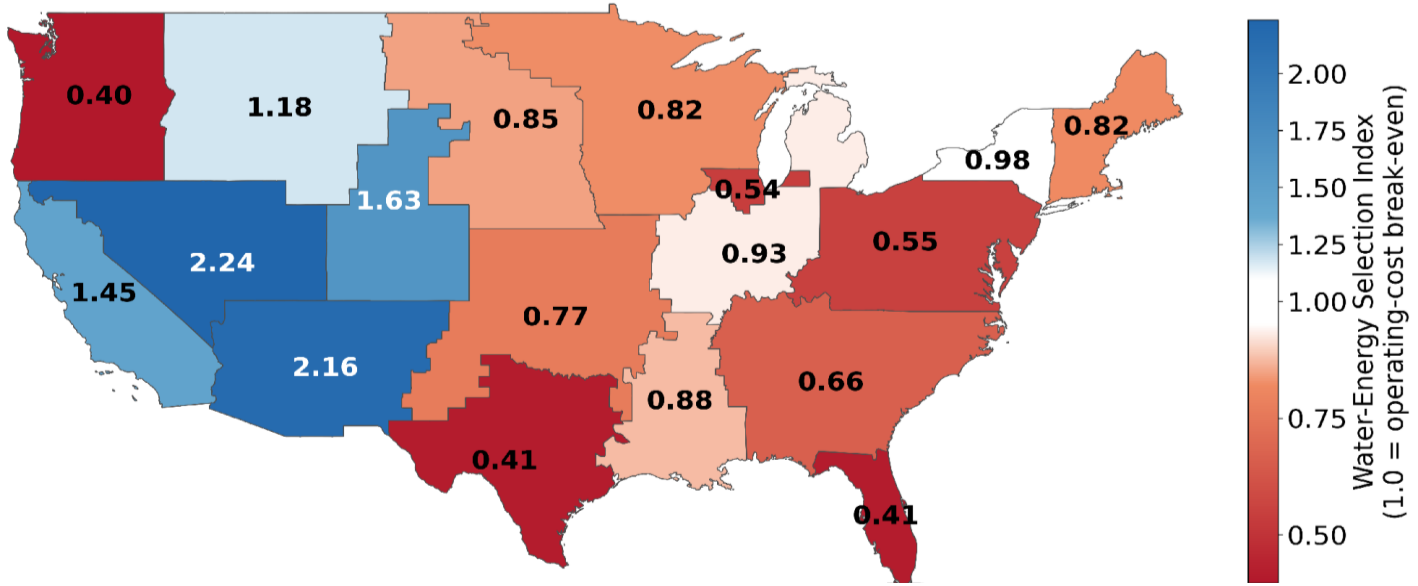
2023 DOE Water + Wastewater Rates

Water-Energy Selection Index (WESI):
When Does Water Pay for Itself?

WESI is a dimensionless Water-Energy Selection Index:

$$\mathbf{WESI = R_{w,BE}/R_w}$$

- **$R_{w,BE}$:** Break-even water rate.
  - *Annual electricity cost saving from wet cooling ÷ additional water use ($/kGal)*
  - *Reflects the maximum water rate justified by the electricity + demand savings from wet cooling*
- **R_w :** Actual regional water rate (same units as $R_{w,BE}$)
- **WESI > 1:** Wet-cooled has lower utility cost
- **WESI < 1:** Dry-cooled has lower utility cost
- **WESI = 1:** Break-even



WESI determines whether the electrical value obtained from evaporation exceeds the cost of that water.

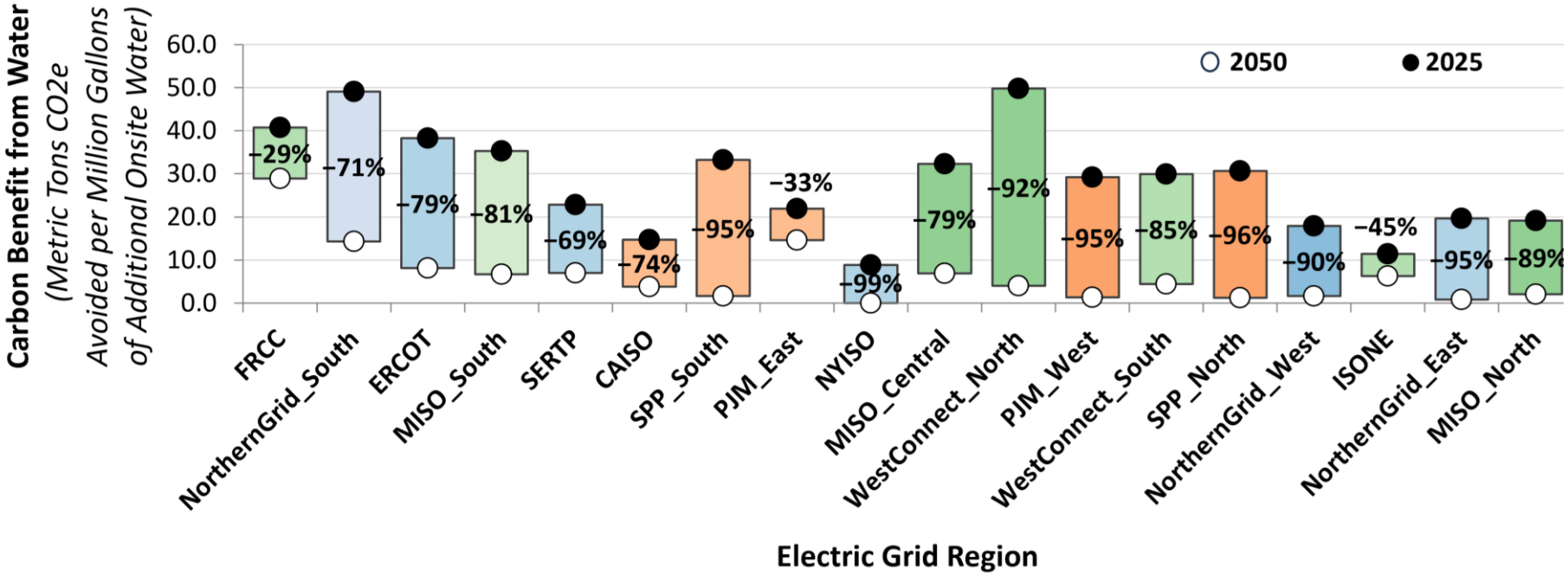
(WESI does not assess water availability or future water stress)

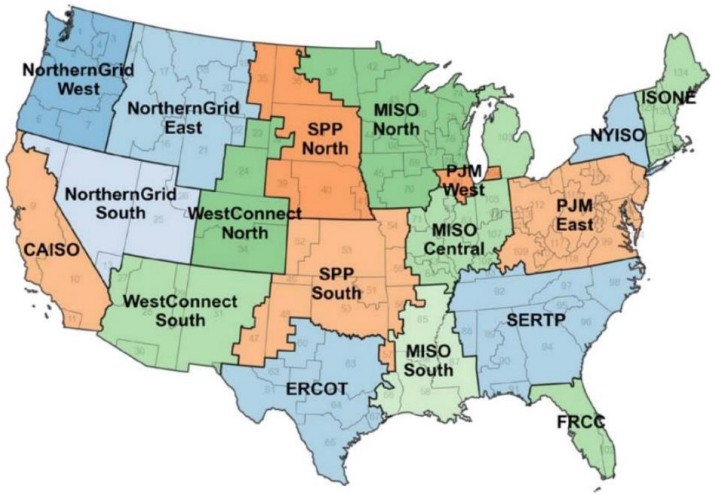
Evaporative cooling is NOT universally cheaper!

The Carbon Benefit of Water Declines as the Grid Decarbonizes

- Metric = **tCO_2e avoided by wet tower per additional Mgal of onsite water relative to dry-cooled**
- Evaporative cooled has lower operational GHG in every modeled region.
- But carbon benefit declines **29–99% by 2050**.

Wet cooling-tower water demand remains unchanged as grids decarbonize, but the carbon value of the electricity it avoids declines.





Electric Grid Regions

Takeaways

1. **Evaporative cooling does not automatically mean lower operating cost.**
   Regional water and electricity economics determine the crossover.

2. **WESI provides a simple regional decision metric.**
   In important markets including PJM East and ERCOT, the model favors dry-cooled on operating cost.

3. **Wet cooling's carbon advantage is not permanent everywhere.**
   Its carbon benefit declines 29–99% by 2050 as grids decarbonize.

2027 ASHRAE Winter Conference Paper expands this framework to dry and hybrid heat rejection and eight total cooling architectures: "*Performance and Sustainability Assessment of Data Center Cooling System Architectures Across U.S. Electrical Grid Regions and Climates"***,** Sammy Houssainy, Jeff Seewald, Karim Shalash

Call to Action: *Stop Treating Cooling Selection as a National Standard*

- Evaluate cooling architecture at the **regional/site level**.
- Include **water + wastewater**, not electricity alone.
- Include **peak-demand charges**, not annual kWh alone.
- Evaluate **future grid emissions**, not only today's carbon intensity.
- Use a crossover metric such as WESI during early design.

The "best" cooling architecture can change by region—and over time.

Thank You!

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